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High quality lamella preparation of gallium nitride compound semiconductor using Triple BeamTM system

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Abstract. Gallium nitride (GaN) compound semiconductors have been known to be very sensitive to Ga focused ion beam (FIB) processing. Due to the nature of GaN based materials it is often difficult to produce damage-free lamellae, therefore applying the Triple BeamTM system which incorporates an enhanced method for amorphous removal is presented to make a high quality lamella. The damage or distortion layer thickness of GaN single crystal prepared with 30 kV Ga FIB and 1 kV Ga FIB were about 17 nm and 1.5 nm respectively. The crystallinity at the uppermost surface remained unaffected when the condition of 1 kV Ar ion milling with the Triple BeamTM system was used. The technique of combining traditional Ga FIB processing with an enhanced method for amorphous layer removal by low energy Ar ion milling allows us to analyse the InGaN/GaN interface using aberration corrected scanning transmission electron microscopy at atomic resolution levels.

1. Introduction

The GaN compound semiconductor is a significant candidate material for light emitted diode circuits or power electronics devices due to its excellent physical properties [1-2]. The electrical performance is strongly related to the nano structure, as a result, the need for analysis of compound semiconductors at the atomic scale using transmission electron microscope (TEM) or scanning transmission electron microscope (STEM) has been of increasing interest. However, it is known that the GaN based compound semiconductor is especially sensitive to the Ga ion of FIB during the lamella production process. A damage layer of several tens of nano meter thickness is formed, degrading the image quality. We have developed the Triple BeamTM system to reduce Ga ion milling related damage. “Triple Beam” is a trademark of Hitachi High-Tech Science Corporation. This technology has been applied to lamella preparation for several types of material such as silicon semiconductor integrated circuits or ceramics. In this study, low damage and high quality lamella preparation condition of GaN compound semiconductor is discussed, and evaluation of the analytical effectivity along the interface at atomic scale resolution.



2. Experimental

The Hitachi NX2000 FIB-SEM Triple Beam™ system was used for sample preparation [3]. Figure 1 shows the diagram of Triple Beam™ system. A FIB column, scanning electron microscope (SEM) column, and Ar/Xe column are installed on the same specimen chamber, designed with each beam to coincide at the same point. The enhanced removal of FIB induced damage using low energy Ar or Xe ion milling is possible in the NX2000 system. A bulk lamella was extracted from the bulk sample using the patented Hitachi micro-sampling in-situ pick up system inside the specimen chamber of the NX2000 system [4].

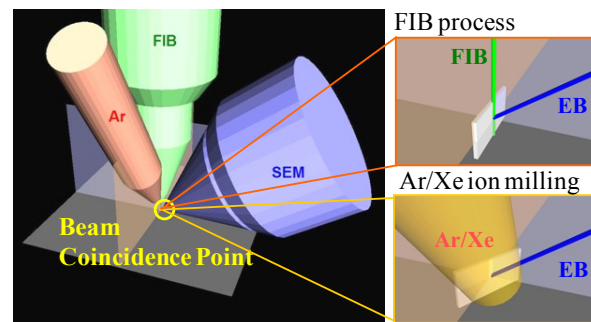


Figure 1. The diagram of Triple Beam™ system.

In the first experiment, the damage layer thicknesses of FIB and ion milling on a GaN single crystal was measured. The damage layer of Ga FIB was formed by acceleration voltage of 30 kV, 10 kV, 5 kV, 1 kV, and finally 1 kV for Ar ion milling. The top surface of the damage layer was protected using electron beam deposition. The five lamellas were prepared to measure each damage layer. An Hitachi HF-3300 cold field emission (FE)-TEM operated at 300 kV was used for the measurement and the evaluation of crystallinity. Optimum lamella preparation condition was determined by those measurement results. In the second experiment, the low damage lamella of nitride semiconductor substrate with InGaN/GaN multi quantum well (MQW) was prepared with optimized sample preparation conditions [5]. The STEM image quality of these lamella was compared. Especially, the sharpness of InGaN/GaN interface was evaluated using an Hitachi HD-2700 aberration corrected STEM and Oxford Instruments X-Max^N100TLE dual silicon drift detector (SDD) system for energy dispersive X-ray (EDX) analysis. The 100 mm² window size and window less detector was adopted for high sensitivity EDX analysis.

3. Results and discussion

Figure 2 shows the high resolution TEM images of GaN single crystal with a formed damage layer intentionally produced. Images showing the results of FIB 30 kV (a), FIB 10 kV (b), FIB 5 kV(c), FIB 1 kV (d) and Ar 1 kV (e), respectively. The electron beam condition for the images is on the $\langle 1\bar{1}00 \rangle$ axial illumination. The diffraction contrast, within the dark area in the TEM images, relates with crystal distortion which was observed in FIB preparation lamella. The thickness of the layer is about 17 nm shown in figure 2(a), which decreased with the decline of FIB acceleration voltage. By contrast, the dark area was removed, revealing the crystallinity of GaN which was kept at the top surface in figure 2(e). It is found that Ar ion milling is effective in removing the FIB damage layer.

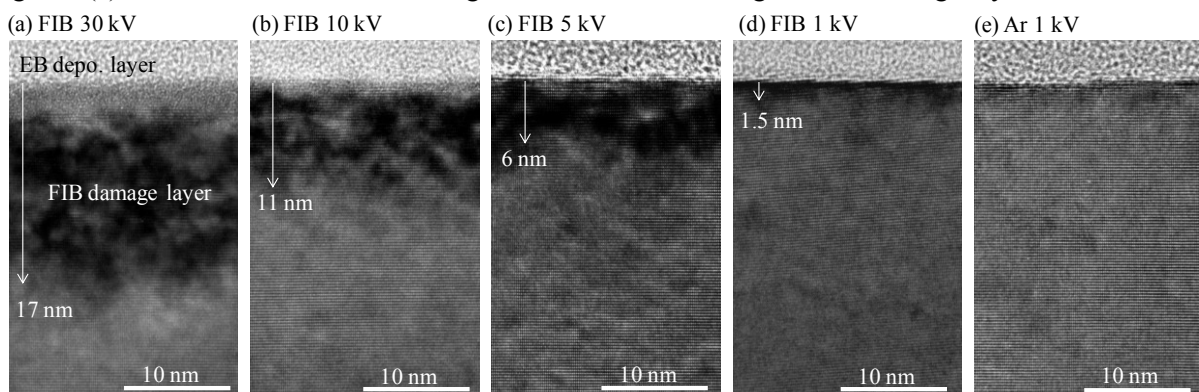


Figure 2. The high resolution TEM images of damage layer of GaN. Data obtained using HF-3300 TEM operated at 300 kV acceleration voltage.

Figure 3 shows the TEM observation result of InGaN/GaN MQW lamella. Images showing the results of FIB 30 kV (a), FIB 5 kV (b), FIB 2 kV (c), Ar 0.5 kV (d) and Xe 1 kV (e), respectively. The incident angle of Ar 0.5 kV is a 20 degree, and Xe 1 kV is a 5 degree. The electron beam incident direction is on the $\langle 11\bar{2}0 \rangle$ axial illumination. The light and shade of TEM image of figure 3(a) is damage, it degrades the image quality. It was reduced with the decline of FIB acceleration voltage, but it was not enough even FIB 2 kV. The results of Ar and Xe milling show clear and smooth TEM image without distortion contrasts, and five MQW layers are clearly observed with high contrast difference. TEM image of Xe milling is smoother than Ar milling. It might be considered that the damage depth of Xe is shallower than Ar, because the mass number of Xe is heavier than Ar. But, due to the difference in milling conditions, further comments are not possible.

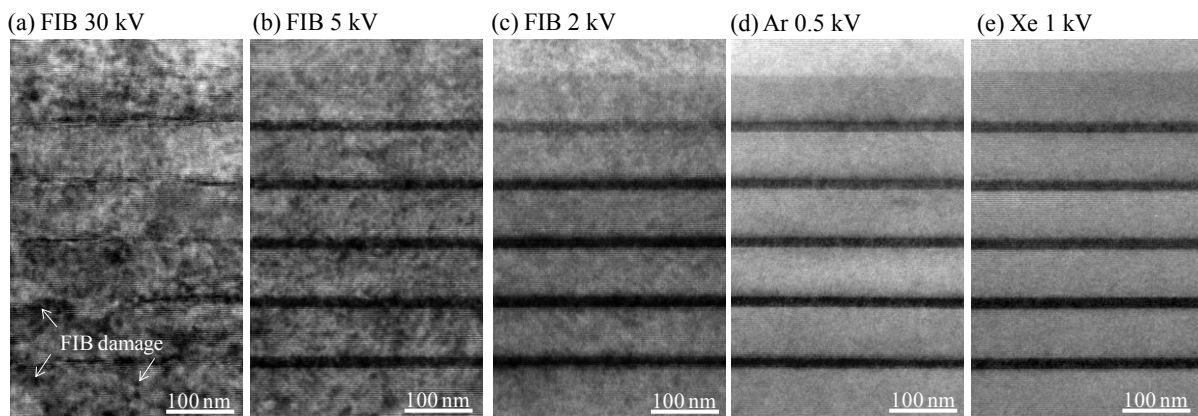


Figure 3. The TEM observation result of InGaN/GaN MQW lamella. These data were taken by using HF-3300 TEM operated at 300 kV acceleration voltage.

Figure 4 shows high resolution annular dark field (ADF)-STEM images of InGaN/GaN MQW lamella processed with Ar 1 kV milling. In figure 4 (a), three layers of InGaN were observed with higher contrast due to the difference of atomic number with GaN. In figure 4 (b), an InGaN layer consisted of eight or nine atomic layers, lower interface of InGaN/GaN is very sharp, but upper interface of GaN/InGaN is rough. The crystal growth condition of the InGaN/GaN MQW contributes to the atomic arrangement of interface, high quality lamella can distinguish this difference.

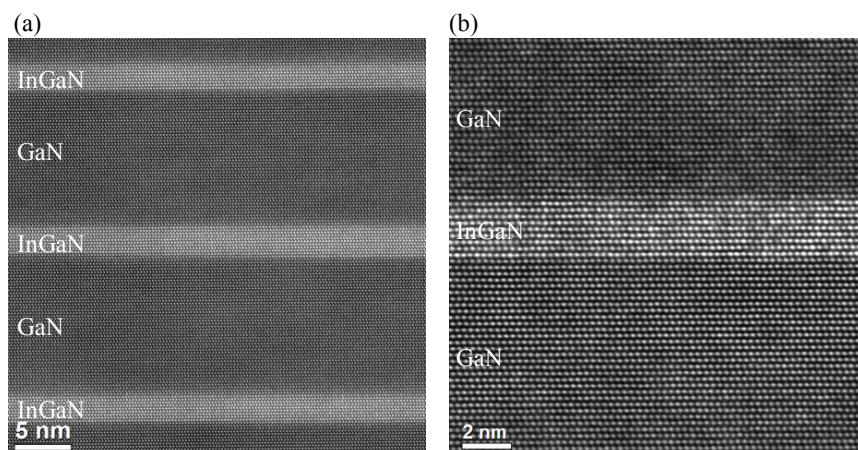


Figure 4. The high resolution ADF-STEM images of InGaN/GaN MQW lamella. 200 kV HD-2700 was used for observation.

Figure 5 shows atomic resolution EDX mapping and line analysis result. Atomic column site of Ga or In were visualized. Line analysis was extracted from EDX spectrum imaging data. It is confirmed

that the InGaN layer consists of eight or nine atomic layers. Distribution of In atom is not uniform in map image, it should be considered that the In atom fluctuation in InGaN layer was detected at atomic resolution scale [6].

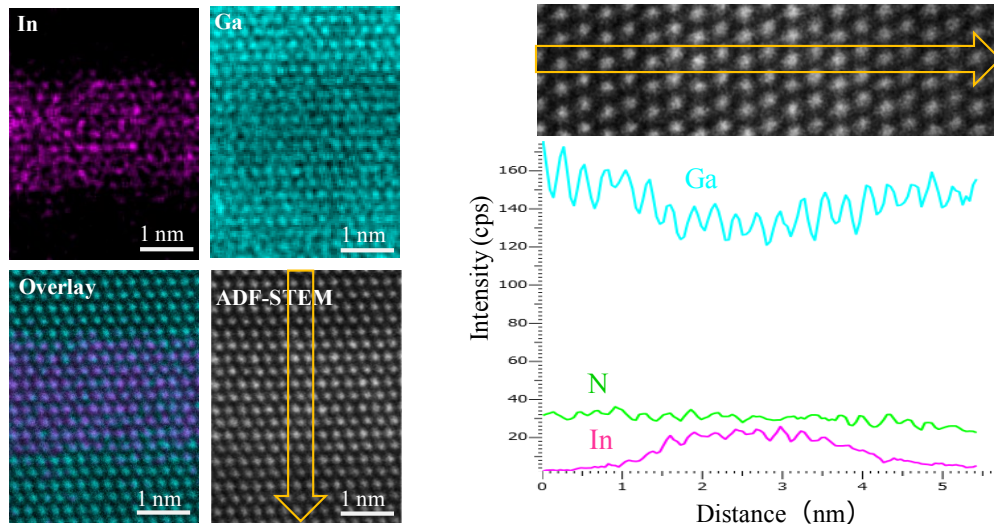


Figure 5. Atomic resolution EDX mapping and line analysis. HD-2700 equipped with X-Max^N 100TLE SDD was used for EDX analysis. Pixel size is 128×100 , acquisition time is about ten minute.

4. Conclusions

We have developed the Triple BeamTM system to remove the Ga induced FIB damage layer and applied it to high quality lamella preparation of GaN compound semiconductor. As a result, the damage layer formed by FIB was perfectly removed by Ar or Xe ion milling, and the TEM image quality of InGaN/GaN MQW lamella was improved, these prepared lamellae allow the analysis at atomic scale resolution levels. It was confirmed that the Triple BeamTM system of NX2000 is a powerful tool to remove Ga induced FIB damage, and it enables the preparation of low damage and high quality lamella of GaN compound semiconductors or other materials.

References

- [1] Amano H, Sawaki N, Akasaki I and Toyoda Y 1986 *Appl. Phys. Lett.* **48** 353-355
- [2] Ohno Y, Kuzuhara M, 2001 *IEEE Trans. on Electron Devices* **48** 517-523
- [3] Takahashi H *et al*, 2006 *Proc. 16th International microscopy congress* p 1104
- [4] Ohnishi T *et al*, 1999 *Proc. 25th int'l Symposium for Testing and Failure Analysis* p 499
- [5] Kaga M, Morita T, Kuwano Y, Yamashita K, Yagi K, Iwaya M, Takeuchi T, Kamiyama S and Akasaki I 2013 *Japanese Journal of Applied Physics* **52**(8) 5208JH07
- [6] Sugahara T, Sato H, Hao M, Naoi Y, Kurai S, Tottori S, Yamashita K, Nishino K, Linda T. Romano and Sakai S 1998 *Japanese Journal of Applied Physics* **37** L1195

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